



18 THE BIGGEST BREAKTHROUGHS YOU WILL SEE IN YOUR LIFETIME

01

3D PRINTED ORGANS

Scientists are using 3D printed tissue, blood vessels and cartilage for surgeries, and the next step is 3D printing organs. The research is focused on hearts, livers and kidneys. People around the world die because of a lack of donors. As 3D printed organs use regenerative cells, there is almost no chance of the body rejecting the organ, which is a bonus.

02

ARCOLOGIES

These are single massive structures with residential, commercial and agricultural zones, that have minimal impact on the environment. Think of them as single buildings that have everything that a city does. Some cities have started using the guiding principles of arcologies, such as the Las Vegas strip.

03

CITIES IN THE SEA

Oceans are potential locations for future cities. We have already seen successful projects in Dubai, Kuwait and Singapore. Future concepts of such cities envision using the space between the seafloor and the surface, for producing energy, harvesting fresh water and growing agricultural produce. MIPL has plans for five cities along India's coastline.

04

DARK MATTER

Dark matter and dark energy could together make

The world has changed spectacularly over the past 80 years, and we are only seeing technological progress accelerating. Here are the changes we can expect to see in a lifetime.

Aditya Madanapalle | aditya@digit.in

up 95 per cent of the known universe. Observations from astronomical instruments clearly show the influence of some unknown material on the expansion of the universe. Scientists have not been able to observe this material yet, hence the name - dark matter. Telescopes dedicated to mapping and studying dark matter might one day solve this long-standing mystery.

05

DE-EXTINCTION

The Pyrenean ibex, a species that went extinct in 1999, was successfully cloned in 2003 using a goat as a surrogate mother, but the cloned ibex died after seven minutes. Still, the effort showed that it was possible to bring back extinct species using cloning. In principle, any species with surviving tissues and a suitable surrogate mother can be brought back from the dead.

06

FLYING CARS

More than a hundred years ago, it was taken for granted that the future would have flying cars for transportation. A number of concepts may make this a reality, although the benefits are not apparent

in terms of cost, fuel economy and time saved for a typical commute distance. There are many companies around the world working on air taxi services.

07

FUSION REACTOR

A fusion reactor is a theoretical device that produces electricity harnessed from the energy released when two atoms combine to form a heavier atom. In conventional nuclear reactors, the energy is provided by nuclear fission, where an atom splits into two smaller atoms. The US and France are at the forefront of the research, using lasers to heat and compress isotopes of hydrogen.

08

HUMAN CONNECTOME

A connectome is a map of all connections in the nervous system. Researchers have successfully mapped the 7000 connections in the 300 neurons of just one worm - a nematode. The human brain has a hundred billion neurons, and is much more difficult to map. Mapping the human connectome is one of those projects that can support a number of other futuristic initiatives, such as the neural lace and mind uploading.